

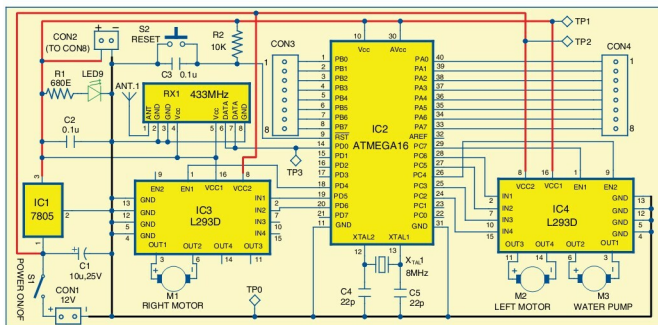


Fig. 4: Circuit diagram of the main module (robot)

TABLE I
Mode Selection Using
DIP1 Switch

PD3	PD4	PD5	PD6	Mode select
1	1	1	1	Manual
0	1	1	1	Reset
1	0	1	1	Autonomous
1	1	0	1	Water pump on
1	1	1	0	LED test mode

EFY Lab.) Different modes of operation are given in Table I. The fire extinguishing robot works in three stages.

Stage 1: Fire detection (autonomous mode). IR photodiodes are connected in reverse bias as shown in the circuit diagram of the sensor module (Fig. 3). Anodes are commonly connected to the ground and cathodes are connected to the 5V via resistors of 1M Ω each.

Voltage across the photodiode is given as input to ADC pins (PA0 through PA7) of ATmega16. When IR waves fall on the IR photodiode, its resistance decreases from 650k Ω to 150k Ω , reducing the voltage across the photodiode, thus changing the input voltage at the ADC pin. By proper quantisation, the presence

and absence of the flame can be distinguished.

Similarly, eight IR photodiodes mounted in a circular fashion on the sensor board help detect the fire; the corresponding LED glows if fire is detected. The cone of detection of the IR photodiode is large, thus decreasing the resolution of the system. This problem can be solved by properly shielding IR photodiodes.

Code to detect the presence of fire using the ADC is as follows:

```
{
unsigned char v;
v= read_adc(0);
if(v>=0 && v <=128) //Stores the digital
value of the analog voltage at ADC 0
fire_detected ( );
if(v>128 && v<=255)
fire_not_detected( );
}
```

Stage 2: Extinguishing fire. Constant feedback from sensors is fed to the main module through CON4, and hence position of the fire with respect to the robot is determined. The main module includes an ATmega16 microcontroller, two L293D motor driver ICs to drive motors, a water pump and RF receiver RX1. The circuit diagram of the main module is shown in Fig. 4.

The basic function of the algorithm is to orient the front sensor in front of the fire so that the nozzle of the pump comes directly above the fire source. When this is achieved, the pump starts and extinguishes the fire.

The robot moves with the help of two motors, whose sense of rotation is controlled by the controller, depending on the feedback from the sensor.

Stage 3: RF communication and manual control (manual mode). The robot is controlled by the operator with the help of a wireless remote (circuit diagram is shown in Fig. 5) that uses an RF module for communicating with the robot.

Switches, push buttons and joysticks are provided on the remote that controls various tasks such as autonomous mode selection, reset and starting the pump. For each command, the remote sends a specific character that is received by the robot and the corresponding operation is performed.

Specimen code for autonomous mode is given as below:

```
For remote (transmitter):
    if(check_bit(SPIND,4)==0)
    {
        i='h'; //auto mode
```